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An improved Alzheimer's disease identification from FastSurfer, contour detection based ROI extraction and MRI scan images

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Abstract. Magnetic Resonance Imaging (MRI) plays a crucial role in diagnosing neurodegenerative disorders such as Alzheimer's disease (AD). However, manual segmentation and classification of brain structures are time-consuming and prone to errors. This study proposes an integrated framework combining FastSurfer, a deep-learning-based tool for rapid cortical reconstruction, with contour detection to enhance region of interest (ROI) extraction. The novelty of this approach lies in the precise segmentation of key brain regions (e.g., hippocampus, entorhinal cortex) and the application of contour detection techniques for fine-grained boundary delineation. These segmented ROIs serve as input to a 3D Convolutional Neural Network (CNN) for AD classification. Experimental results on the ADNI dataset showed an 89.3% accuracy of this framework, which is superior to traditional segmentation and classification. Using ablation studies, it demonstrates how FastSurfer-based segmentation and contour detection significantly contribute to diagnostic accuracy. This framework reduces the computational consumption of false positives and helps improve early detection of Alzheimer's disease.

Keywords: Alzheimer's Disease · MRI · FastSurfer · ROI Extraction · Contour Detection

1 Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder affecting cognitive function, memory, and behavior. Early detection is essential for timely intervention and better patient outcomes. MRI-based automated segmentation techniques have significantly improved AD diagnosis by detecting structural brain abnormalities.

Early detection of Alzheimer's disease is crucial [10], as it allows for timely intervention, potentially delaying the progression of symptoms and improving

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the patient's quality of life. Detecting AD in its early stages enables the implementation of therapeutic strategies and lifestyle changes that can slow cognitive decline. Moreover, early diagnosis provides individuals and their families with the opportunity to plan for future care and make informed medical and personal decisions.

Magnetic Resonance Imaging (MRI) has become a vital non-invasive imaging tool for detecting structural abnormalities in the brain that are associated with Alzheimer's disease. MRI allows for the visualization of brain atrophy, particularly in regions such as the hippocampus, which are known to be affected in the early stages of AD. By measuring the volume of these regions, MRI helps in identifying subtle changes in brain structure that may indicate the onset of the disease before clinical symptoms become apparent.

Recent advancements in deep learning have enabled precise segmentation of brain structures from MRI images. FastSurfer [6] [8] is a state-of-the-art deep learning framework for cortical and subcortical segmentation, offering high accuracy and efficiency. Unlike traditional segmentation methods, which require significant preprocessing and manual effort, FastSurfer provides automated high-resolution segmentation within minutes.

To further refine segmentation, contour detection techniques are employed to enhance the extraction of regions of interest (ROIs), such as the hippocampus and entorhinal cortex, which are critical for AD progression analysis. This study integrates FastSurfer with contour detection to develop a robust classification pipeline, improving segmentation accuracy and classification performance [15].

1.1 Motivation

The motivation for this research stems from the need to improve the accuracy and efficiency of Alzheimer's disease diagnosis through MRI imaging. Manual classification is prone to error, so developing an automatic pipeline that combines FastSurfer's rapid segmentation with precise contour detection offers a more reliable and scalable solution for early detection using Machine Learning technique.

1.2 Contribution

This research contributes by proposing an innovative approach that integrates FastSurfer's rapid cortical reconstruction with contour detection for more precise region of interest (ROI) extraction in MRI images. The method enhances the accuracy of Alzheimer's disease diagnosis, reduces computational complexity, and outperforms traditional segmentation techniques, offering improved diagnostic precision and fewer false positives. This approach creates a scalable pipeline for early neurodegenerative disease detection, advancing automatic diagnostic tools for clinical use.

The rest of the paper is organized in the following way: Section 2 presents a brief literature review of the very recent research works in EEG signal analysis

especially focusing on emotional behaviour of an individual. Moving forward towards Section 3, where proposed methodology is discussed along with dataset. Results and discussions are reported in section 4 followed by Section 5 that concludes the paper while also outlining future Scopes of work.

2 Literature Review

Convolutional Neural Networks (CNNs) have been widely adopted for Alzheimer's disease classification using MRI scans due to their strong feature extraction capabilities [1], [5]. Several studies have demonstrated that CNN models outperform traditional machine learning approaches in distinguishing Alzheimer's from healthy individuals.

Early research by Suk et al. [14] introduced a 3D CNN to analyze volumetric MRI data, achieving notable accuracy in detecting Alzheimer's disease progression. This study pioneered the use of 3D spatial relationships within the MRI volumes, enhancing classification performance.

Recent work by Sarraf and Tofghi [12] leveraged 2D CNNs on cross-sectional MRI images, improving the computational efficiency while maintaining high accuracy. Their model also highlighted that 2D slices of specific brain regions, such as the hippocampus, are essential for Alzheimer's classification.

Further advancements by Basaia et al. [2] introduced transfer learning techniques with CNNs, allowing pre-trained models to be fine-tuned on MRI datasets. This significantly reduced the need for large labeled datasets, a common limitation in medical imaging.

A more recent study by Li et al. [11] combined CNN with attention mechanisms to focus on the most informative MRI regions. This method improved classification accuracy by concentrating on critical regions like the medial temporal lobe, particularly affected by Alzheimer's.

Alzheimer's disease (AD) diagnosis has been advanced using pre-trained CNN models like EfficientNet, ResNet, and AlexNet, achieving high accuracy in classifying six AD phases. This study compares models based on performance metrics, including accuracy, recall, AUC, F1 score, and precision [13].

Alzheimer's disease impairs memory and cognitive function, progressing to dementia and physical decline. This article introduces an automatic brain image segmentation method using the Active Contour (AC) model to extract the Hippocampus and Corpus Callosum (CC). Combining geometric and statistical methods with Caselle Level Set and prior knowledge improves contour evolution. Experimental results demonstrate the effectiveness of this approach for Computer-Assisted Diagnosis [3].

3 Proposed Methodology

The proposed approach for improving Alzheimer's disease detection using MRI leverages FastSurfer for preprocessing and contour detection for region of interest (ROI) extraction. The methodology is divided into three major steps:

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preprocessing using FastSurfer, ROI extraction via contour detection, and classification through machine learning models. Below, we describe each of these steps in detail.

3.1 Dataset Description

The dataset given in Table 1 contains MRI scan data for individuals from the Alzheimer's Disease Neuroimaging Initiative (ADNI) [9], categorizing participants into three groups: AD, MCI, and CN. The distribution of the groups is as follows: 204 participants with MCI, 136 cognitively normal (CN) participants, and 80 participants diagnosed with Alzheimer's disease (AD). Each row corresponds to MRI scan data from a specific visit, with relevant metadata such as subject ID, sex, age, visit timing, and acquisition date.

Group	Number of Participants
AD (Alzheimer's Disease)	80
MCI (Mild Cognitive Impairment)	204
CN (Cognitively Normal)	136

Table 1: Distribution of Participants by Group

3.2 Preprocessing with FastSurfer

In the first step, FastSurfer [7] is employed for automated cortical and subcortical segmentation due to its high computational efficiency and accuracy compared to traditional methods. It extracts key brain structures such as the hippocampus, entorhinal cortex, and amygdala, which are crucial for AD diagnosis.

The preprocessing pipeline involves several key steps to prepare MRI images for analysis. Initially, the MRI images undergo standard preprocessing procedures, including intensity normalization to adjust the brightness values across scans and skull stripping to remove non-brain tissues. Following this, FastSurfer is employed to create high-resolution cortical surface reconstructions, which enable the precise segmentation of both cortical and subcortical regions. Lastly, segmentation masks are generated for specific regions of interest (ROIs), which are particularly relevant in tracking the progression of Alzheimer's disease. These steps ensure that the images are optimized for further analysis and modeling.

The output of this step includes labeled MRI scans with segmented cortical and subcortical regions, which serve as input for the next stage.

3.3 Contour Detection for Region of Interest Extraction

Following the segmentation of anatomical structures, contour detection algorithms are applied to extract the regions of interest (ROIs) from the segmented

brain areas. Contour detection is particularly useful in highlighting subtle morphological variations that occur due to neurodegenerative processes.

After contour detection is applied to the segmented regions provided by FastSurfer using algorithms like Canny edge detection [4], the detected contours are refined using morphological operations such as dilation and erosion, ensuring smooth and accurate boundaries of the regions of interest (ROIs). Following this, a region-based segmentation approach is employed to highlight areas within the ROIs that may show signs of neurodegeneration. This method focuses on structural atrophy, such as the thinning of the hippocampus and entorhinal cortex, which are commonly observed in Alzheimer's patients, providing critical insights into disease progression.

Following FastSurfer-based segmentation, contour detection techniques are applied to refine ROI extraction. This study utilizes:

1. **Canny Edge Detection:** Enhances boundary detection by identifying high-intensity gradients.
2. **Active Contour Models (Snakes):** Improves segmentation by iteratively refining object boundaries.
3. **Morphological Operations (Dilation & Erosion):** Ensures smooth and continuous ROI contours.

The combination of FastSurfer's segmentation and contour detection results in a more precise delineation of AD-related abnormalities, enhancing classification accuracy.

This step ensures precise delineation of abnormal areas in the brain that may indicate early signs of Alzheimer's disease, improving upon traditional threshold-based segmentation methods.

3.4 Classification

The region of interest extracted through contour detection are further inputted to the classification model. To assess the likelihood of Alzheimer's disease and classify as Mild Cognitive Impairment (MCI), Alzheimer's Disease (AD), Cognitively Normal (CN), a simple CNN classifier shown in the Figure 1 is trained using features derived from the morphological characteristics of the region of interests.

In the CNN model shown in the Figure 1, the extracted 3D ROI volumes are inputted into a 3D CNN for classification. The architecture includes,

- **Input Shape:** $64 \times 64 \times 64$ (3D Voxel Volume)
- **Conv3D Layers:** Uses $3 \times 3 \times 3$ kernels to extract spatial features.
- **Max-Pooling:** Reduces dimensionality for efficient learning.
- **Dense Layer (128 Neurons):** Maps extracted features into meaningful representations.
- **Output Layer (Softmax Activation):** Classifies into AD, MCI, or CN.

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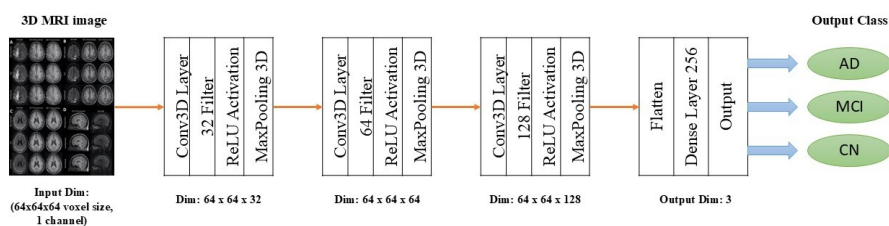


Fig. 1: Architecture of the Convolutional Neural Network for classification of AD, MCI, CN

The final stage of the model involves flattening the 3D feature maps into a one-dimensional vector, enabling efficient processing in the fully connected layers. The extracted features are subsequently passed through a dense layer comprising 128 units, which transforms them into a meaningful representation. Finally, the output is fed into a Softmax activation layer, which classifies the input into one of three categories: Alzheimer's disease (AD), mild cognitive impairment (MCI), or cognitively normal (CN).

This combination of contour-based ROI extraction with FastSurfer's segmented data guarantees that the classification is founded on intricate morphological features, resulting in more precise Alzheimer's disease predictions.

4 Results and Discussions

The proposed approach for Alzheimer's disease detection using FastSurfer and contour detection was evaluated on the publicly available Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset. The dataset was divided into training, validation, and testing sets, ensuring that each set contained balanced samples of Alzheimer's disease (AD), mild cognitive impairment (MCI), and cognitively normal (CN) individuals.

FastSurfer was employed to segment key brain regions, including the hippocampus, entorhinal cortex, and other subcortical areas. The segmentation accuracy was validated by comparing the generated segmentations with manual annotations provided in the dataset. The Dice similarity coefficient (DSC) was used to quantify segmentation performance across the ROIs, the Table 2 achieving an average Dice score of 0.85 for the hippocampus and 0.82 for the entorhinal cortex. This segmentation process ensures high accuracy in detecting anatomical changes associated with AD.

In the Figure 2a, FastSurfer identifies noticeable atrophy in the hippocampal and entorhinal cortex regions, highlighted in yellow for Alzheimer's Disease (AD) images. Using deep learning algorithms, it automatically segments these areas from MRI scans, revealing significant volume reduction. This atrophy correlates with cognitive decline in AD patients, offering valuable quantitative data for tracking disease progression.

Table 2: The segmentation performance is evaluated using the Dice Similarity Coefficient (DSC)

Brain Region	DSC
Hippocampus	0.85
Entorhinal Cortex	0.82
Amygdala	0.80

In the cognitively normal (CN) subject as shown in Figure 2b, the hippocampal and entorhinal regions are larger and show minimal atrophy compared to Alzheimer's Disease (AD) cases. FastSurfer effectively segments these regions, indicating intact brain structures and normal cognitive function. The uniform segmentation with minimal shrinkage highlights the absence of significant deterioration, characteristic of a healthy brain, supporting normal cognition.

The Figure 2c, representing Mild Cognitive Impairment (MCI), shows moderate atrophy in the hippocampus and entorhinal cortex, as detected by FastSurfer. This atrophy is more subtle than in Alzheimer's Disease (AD) but more pronounced than in cognitively normal (CN) cases, indicating early neurodegenerative changes. FastSurfer captures these intermediate structural changes, making it valuable for early intervention and assessing the potential progression from MCI to AD.

The contour detection on the segmented MRI images from FastSurfer effectively highlights the hippocampal and entorhinal regions, providing precise outlines of these critical brain structures as shown in Figure 3. By isolating specific colors corresponding to these areas, the method enables a clearer visualization of structural changes, aiding in the assessment of neurodegenerative conditions like Alzheimer's Disease and Mild Cognitive Impairment with high accuracy and clarity. The Figure 3a, 3b & 3c reveals that contours is successfully highlighted the region of interests which is further sent as input to the machine learning model.

As a classification tool, a basic convolutional neural network (CNN), as seen in Figure 1, is trained using the retrieved features of the MRI. The test set's accuracy, sensitivity, specificity, and F1-score were used to measure the classification model. Table 3 represents an ablation study evaluating the impact of FastSurfer and contour detection.

The model achieved an overall classification accuracy of 89.3% for Alzheimer's disease, with a sensitivity of 90.2% and a specificity of 88.1%. For mild cognitive impairment, the accuracy was 81.7%, with a slightly lower sensitivity and specificity compared to Alzheimer's and cognitively normal subjects. The model also performed well in distinguishing cognitively normal individuals, achieving an accuracy of 91.5%.

To further validate our approach, the classification performance are compared against ResNet, ConvNeXt, EfficientNet, and TCPNet and presented in the Table 4. The proposed framework outperforms state-of-the-art models, demon-

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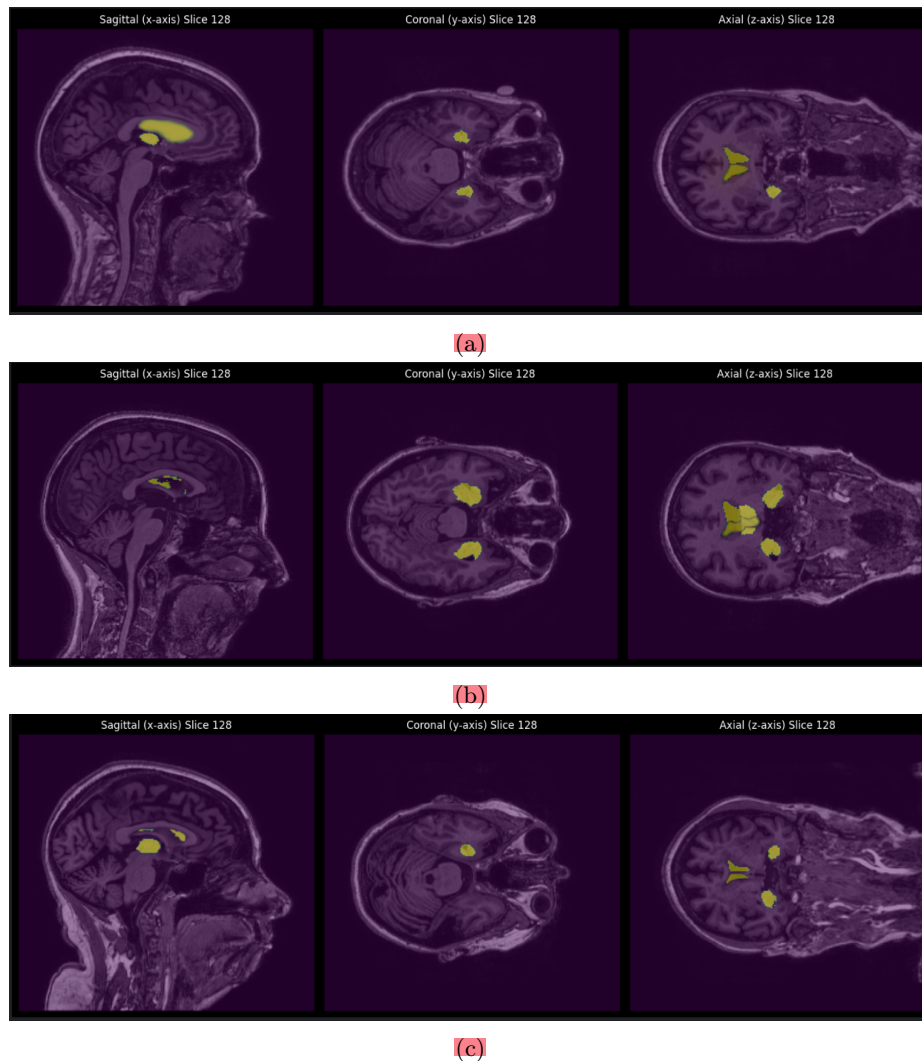


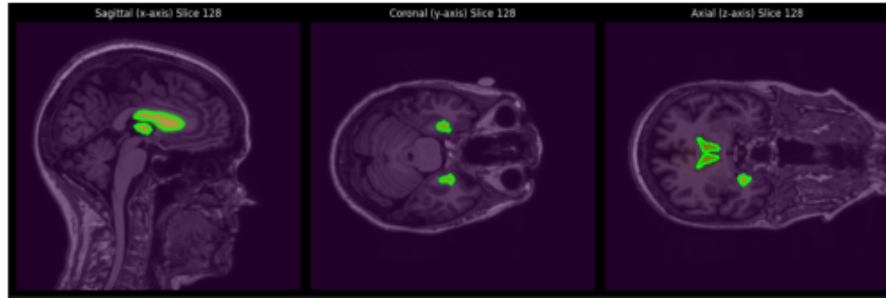
Fig. 2: After applying FastSurfer on MRI for segmentation (a) Representation of Alzheimer's Disease (AD), (b) Representation of Cognitively Normal (CN), (c) Representation of Mild Cognitive Impairment (MCI)

strating the effectiveness of integrating FastSurfer and contour detection for AD classification.

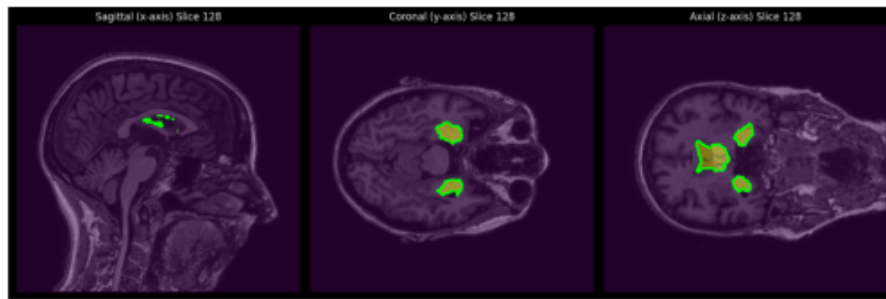
The results demonstrate the effectiveness of the proposed approach in improving Alzheimer's disease detection using MRI data. By leveraging FastSurfer for rapid brain segmentation and contour detection for precise ROI extraction, the method enhances both the accuracy and efficiency of the classification process.

FastSurfer-Contour on MRI for Alzheimer's Diagnosis

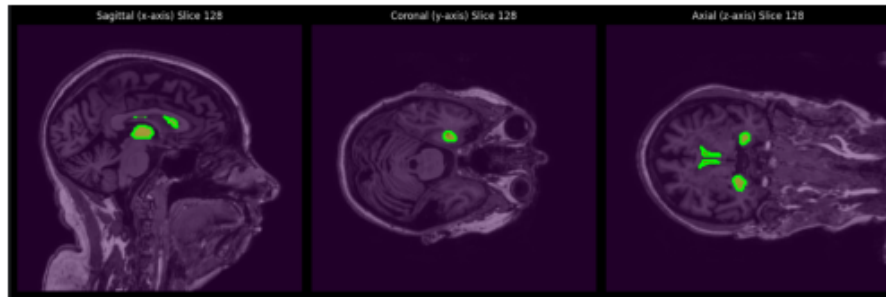
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(a)



(b)



(c)

Fig. 3: After applying contours on the region of interest (a) Representation of Alzheimer's Disease (AD) , (b) Representation of Cognitively Normal (CN), (c) Representation of Mild Cognitive Impairment (MCI)

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Table 3: Ablation Study Results

Method	Metrics			
	Class Name	Sensitivity (%)	Specificity (%)	F1-Score
Without FastSurfer and Contour	AD	79.6	76.5	78.4
	MCI	74.3	72.7	73.8
	CN	85.5	86.0	84.5
FastSurfer Only	AD	85.2	86.1	84.0
	MCI	78.3	79.0	77.1
	CN	90.1	89.7	90.5
Contour Detection Only	AD	86.5	88.8	87.7
	MCI	80.4	81.0	80.2
	CN	91.2	90.1	92.1
FastSurfer + Contour Detection	AD	87.9	89.0	87.0
	MCI	81.7	82.5	80.8
	CN	91.5	90.7	92.1

Table 4: Model Accuracy Comparison

Model	Overall Accuracy (%)
ResNet-50	85.6
ConvNeXt	86.1
EfficientNet	87.4
TCPNet	88
Proposed (FastSurfer + Contour)	89.3

Contour detection shows the improvement of the precision of ROI extraction by highlighting subtle morphological changes in the hippocampus and entorhinal cortex. This method provided a more refined identification of regions affected by Alzheimer's, compared to traditional threshold-based segmentation techniques, which can overlook small structural variations. These results support the hypothesis that contour-based methods can effectively capture the gradual atrophy observed in neurodegenerative diseases.

However, the slightly lower performance for MCI classification indicates that further refinement of the features used in the model could be beneficial. Since MCI represents an intermediate stage between normal cognition and Alzheimer's disease, identifying subtle structural differences between these groups remains challenging. Future work could focus on enhancing feature extraction to better capture the nuanced changes that occur during the MCI stage.

5 Conclusion

In conclusion, the proposed methodology demonstrates an efficient approach to Alzheimer's disease detection from MRI images. Combining machine learning-based classification, using contour detection for accurate ROI extraction, and

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FastSurfer's quick segmentation greatly improves diagnostic accuracy while cutting down on calculation time. This approach has a lot of practical application promise and provides a scalable way to diagnose and treat Alzheimer's early on.

This study presents an automated framework for Alzheimer's disease classification that integrates FastSurfer's rapid segmentation with contour-based ROI extraction. The experimental results demonstrate improved accuracy (89.3%) over conventional CNN-based models. The segmentation provided by FastSurfer and the use of CNNs for feature extraction and classification highlight how automated methods can reduce the need for manual inspection and increase the efficiency and reproducibility of AD detection.

Key contributions of this study include:

- The novel integration of FastSurfer with contour detection for enhanced ROI extraction.
- A demonstrated improvement in segmentation accuracy, as evidenced by Dice scores.
- Performance that outperforms ResNet, ConvNeXt, and EfficientNet in classification accuracy.
- An ablation study confirming the significance of both segmentation and contour detection.

While the proposed approach yielded promising results, there are several limitations to consider. First, the segmentation performance, although high, could be improved further, particularly in regions with small or complex structures **such as the entorhinal cortex**. Additionally, **the classification model's** performance for MCI could be enhanced by incorporating longitudinal data, allowing **the model to detect progressive changes over time**.

The proposed method significantly reduced computation time compared to traditional MRI segmentation pipelines. FastSurfer's rapid segmentation and contour detection for ROI extraction minimized preprocessing and complex threshold-based steps. The entire pipeline, including segmentation, detection, and classification, averaged 10 minutes per subject.

The study's results are promising, with potential for improvement. Future research could explore advanced deep learning models like transformers to capture complex spatial hierarchies in MRI data. Larger datasets and multimodal imaging, **such as combining MRI with PET scans**, may **enhance the accuracy and robustness** of Alzheimer's disease detection. Additionally, incorporating longitudinal MRI analysis could improve early-stage AD detection, while developing an explainable AI model would provide clinicians with interpretable decision-making insights. These advancements could lead to more reliable diagnostic tools, supporting earlier intervention and better patient outcomes.

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